Note on Commutative Algebra

Jiawen Huang

22 January 2026

Contents

Acknowledgement

This note is basically a copy of Eisenbud's Commutative Algebra with a View Toward Algebraic Geometry, with some additional explanations and examples from Eisenbud's lectures and my own understanding.
This latex style is made by Evan Chen.

January 17, 2026

Corollary 0.1 (Summand of polynomial rings over fields is finitely generated)
Let k be a field, and let R be a k-subalgebra of $S = k[x_1, \ldots, x_n]$. R is a direct summand of S in the sense that there is a surjective R-algebra homomorphism $\phi : S \to R$ such that $\phi|_R = \mathrm{id}_R$ and ϕ preserves degrees. Then R is finitely generated as a k-algebra.

Proof. Intuition: R itself is a graded k-algebra, with grading inherited from S. As a graded algebra, R can be decomposed into its homogeneous components. R_0 is easy to get, just include 1 in the generators. Therefore, to generate R, we really need to generate the positive degree part. That motivates us to look at the ideal R_+.

But R_+ may not be finitely generated as an ideal. However, we know that S is a Noetherian ring, so we can look at the ideal R_+S. It must be finitely generated, say by $f_1, \ldots, f_m \in R_+$.

Now, we claim that $R = k[1, f_1, \ldots, f_m]$. Certainly, the right hand side is contained in the left hand side. For the other direction, it suffices to show that any homogeneous element of R can be generated by $f_1, \ldots, f_m$ with coefficients in k. We prove this by induction on degree d. $d = 0$ is trivial. Now suppose we have shown this for degree less than d. Take any homogeneous element $r \in R$ of degree d. Since $r \in R_+S$, we can write $r = \sum g_i f_i$ for some homogeneous element $g_i \in S$. Apply ϕ to both sides, we get

$$r = \phi(r) = \sum \phi(g_i)\phi(f_i) = \sum \phi(g_i) f_i.$$

Note that $\deg(\phi(g_i)) = \deg(g_i) < d$, so by the inductive hypothesis, each $\phi(g_i)$ can be generated by $f_1, \ldots, f_m$ over k. Therefore, r can also be generated by $f_1, \ldots, f_m$ over k. This finishes the proof. □

Remark 0.2. This shows that the invariant ring of a linearly reductive group acting on a polynomial ring over a field is finitely generated. If the group G is finite, the surjective homomorphism ϕ can be taken as $\phi(f) = \sum_{g \in G} gf$. (Weyl's averaging trick)

Remark 0.3. Graded rings are great because it allows us to use induction on degree. So we only need to worry about homogeneous elements.

January 20, 2026

First day of Commutative Algebra course!

We discuss the history of Commutative Algebra and three of the four big theorems.

Theorem 0.4 (Hilbert's Basis Theorem)
If R is a Noetherian ring, then the polynomial ring $R[x]$ is also Noetherian.

Proof. Let I be an ideal of $R[x]$. Our goal is to show that I is finitely generated. We first pick the smallest degree nonzero polynomial f_1 in I. Next, we take f_2 to be the smallest degree polynomial in I which is not in the ideal generated by f_1, that is $f_2 \in I \setminus (f_1)$. Continuing this process, we get a sequence of polynomials $f_1, f_2, \ldots$ in I such that f_n is the smallest degree polynomial in $I \setminus (f_1, \ldots, f_{n-1})$. Clearly we have $\deg(f_{i+1}) \geq \deg(f_i)$ for all i. Since R is Noetherian, the leading coefficients of f_i generate a finitely generated ideal in R. This means that there exists j such that the leading coefficient of f_1 to f_j generate all the leading coefficients of f_i for $i > j$.

Now, we look at f_{j+1}. Its leading coefficient can be generated by the leading coefficients of f_1 to f_j, that is $a_{j+1} = \sum_{i=1}^{j} r_i a_i$ for some $r_i \in R$. Consider the polynomial

$$g = f_{j+1} - \sum_{i=1}^{j} r_i x^{\deg(f_{j+1}) - \deg(f_i)} f_i$$

Since $\deg(g) < \deg(f_{j+1})$ so $g \in (f_1, \ldots, f_j)$. Thus, $f_{j+1} \in (f_1, \ldots, f_j)$, contradicting the choice of f_{j+1}. Therefore, $I = (f_1, \ldots, f_j)$ is finitely generated. □

Theorem 0.5 (Hilbert's syzygy theorem)
If $R = k[x_1, \ldots, x_n]$ is a graded polynomial ring over a field k, then every finitely generated graded R-module $M = \bigoplus_{s \in \mathbb{Z}} M_s$ has a finite graded free resolution of length at most n. That is, the sequence

$$0 \to F_n \to F_{n-1} \to \cdots \to F_1 \to F_0 \to M \to 0,$$

where each F_i is a finitely generated graded free R-module, is exact.

Proof is omitted.

Theorem 0.6 (Hilber's function theorem)

Let M be a finitely generated graded module over the polynomial ring $R = k[x_1, \ldots, x_n]$. Then there exists a polynomial $P(t) \in \mathbb{Q}[t]$ such that for all sufficiently large d, we have $\dim_k M_d = H_M(d) = P(d)$.

We will present two proofs of this theorem.

Proof. We will use the Hilbert's syzygy theorem to prove this. By the theorem, we have a finite graded free resolution of M:

$$0 \to F_n \to F_{n-1} \to \cdots \to F_1 \to F_0 \to M \to 0.$$

Thus, we have $H_M(d) = \sum_{i=0}^{n}(-1)^i H_{F_i}(d)$. So we only need to study the dimension of free modules F_i, which is a direct sum of $k[x_1, \ldots, x_n]$. Note that $H_R(d) = \dim_k R_d = \binom{n+d-1}{n-1}$ (star and bar counting trick), which is a polynomial in d when d is sufficiently large. Therefore, $H_M(d)$ is also a polynomial in d for sufficiently large d. $\square$

Remark 0.7 (The Euler characteristic)**.** Why does $H_M(d) = \sum_{i=0}^{n}(-1)^i H_{F_i}(d)$ hold? This result is a property of the Euler characteristic. For any complex of finite-dimensional vector spaces

$$V_* : 0 \to V_n \to V_{n-1} \to \cdots \to V_1 \to V_0 \to 0,$$

We define its Euler characteristic to be

$$\chi(V_*) = \sum_{i=0}^{n}(-1)^i \dim_k V_i.$$

There is a nontrivial fact that the Euler characteristic is 0 if the complex is exact. The easiest proof is via the decomposition of exact sequence into short exact sequences and sum up the dimension relations for each short exact sequence.

$$\begin{tikzcd}
 & & 0 \arrow[dr] & & 0 & & \\
 & & & \operatorname{im} d_{i+1} = \ker d_i \arrow[ur] \arrow[dr] & & & \\
M_{i+2} \arrow[rr, "d_{i+2}"] \arrow[dr] & & M_{i+1} \arrow[rr, "d_{i+1}"'] \arrow[ur] & & M_i \arrow[rr, "d_i"] \arrow[dr] & & M_{i-1} \\
 & \operatorname{im} d_{i+2} = \ker d_{i+1} \arrow[ur] \arrow[dr] & & & & \operatorname{im} d_i = \ker d_{i-1} \arrow[ur] \arrow[dr] & \\
0 \arrow[ur] & & 0 & & 0 \arrow[ur] & & 0
\end{tikzcd}$$

Second one is via induction on n.

Proof. $n = 0$ is trivial. Suppose it's true for number less than n. Consider the short exact sequence

$$0 \to (0 : x_n) \to M(-1) \xrightarrow{x_n} M \to M/x_nM \to 0.$$

Notice that M/x_nM is a finitely generated graded module over $k[x_1, \ldots, x_{n-1}]$ (x_n acts as zero) and $(0 : x_n)$ is also a finitely generated graded module over $k[x_1, \ldots, x_{n-1}]$ (the generators of $(0 : x_n)$ over $k[x_1, \ldots, x_{n-1}, x_n]$ also generate $(0 : x_n)$ over $k[x_1, \ldots, x_{n-1}]$ since x_n would act as zero). By the inductive hypothesis, both M/x_nM and $(0 : x_n)$ have Hilbert dimensions agree with polynomials when d is large enough. Thus, by the exactness of the sequence,

$$H_M(d) - H_M(d-1) = H_{M/x_nM}(d) - H_{(0:x_n)}(d-1) = \lambda(d).$$

for some polynomial $\lambda(d)$ when d is large enough. Thus,

$$H_M(d) - H_M(0) = \sum_{i=1}^{d}(H_M(i) - H_M(i-1)) = \sum_{i=1}^{d} \lambda(i).$$

Notice that $\sum_{i=1}^{d} i^k$ is a polynomial in d of degree $k+1$. Therefore, $\sum_{i=1}^{d} \lambda(i)$ is also a polynomial in d when d is large enough. Therefore, $H_M(d)$ also agrees with a polynomial when d is large enough. □

Remark 0.8. Is $H_M(d)$ always finite? The answer is yes if M is a finitely generated module over $k[x_1, \ldots, x_n]$. This is nontrivial because $H_M(d)$ is considering the dimension of M as module over k, not over R.

Proof. Consider $M' = \bigoplus_{s=d}^{\infty} M_s$. M' is a submodule of M, so it's finitely generated as R-module. Let $m_1, \ldots, m_t$ be the generators of M'. Each m_i has degree at least d. So for any element in M_d, it's a linear combination of m_i with coefficients in R of degree at most 0 (otherwise it would have degree greater than d). Therefore, M_d is spanned by $m_1, \ldots, m_t$ over k, which is finite-dimensional. □

I came up with another more intuitive proof.

Proof. M is finitely generated as R-module by $m_1, \ldots, m_t$. For some degree d, any element in M_d must be a linear combination of m_i with coefficients as the monomials in R of degree at most $d - \deg(m_i)$. There are only finitely many such monomials, so M_d is finite-dimensional over k. □

January 22, 2026

Today we discuss the Hilbert's Nullstellensatz and the equivalence between the category of affine algebraic varieties and the category of finitely generated reduced k-algebras. For further details, please refer to here.